

THINURA KAHADUWA

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PROFILE

AI-focused IT undergraduate with production internship experience on a multi-tenant document digitization platform: LLM-driven information extraction, OCR pipelines, and embedding-based entity resolution in Python and FastAPI. Comfortable owning a feature end to end: data model, migrations, async pipeline steps, performance profiling, and testing. Seeking an Associate AI Engineer role applying AI and NLP to real, messy data.

EXPERIENCE

AI Engineering Intern — Embla Software Innovation (Pvt) Ltd, Colombo, Sri Lanka 02/2026 – 08/2026
GeneXtract — multi-tenant document digitization platform for historical church records (computer vision, OCR, NLP)

- Turned unstructured OCR text into queryable location data with LLM-driven extraction and normalization, adding provenance-tracked confidence scores and deduplication so records could be trusted downstream.
- Enabled cross-document identity linking by implementing entity resolution with 768-dimensional multilingual sentence embeddings and pgvector similarity search, surfacing duplicate person records for merge; kept the pipeline responsive by moving embedding inference off the async event loop.
- Owned the parsing feature set: parent-name parser and component extraction, person-to-family relationship matching, and a date-of-birth parser overhaul with idempotent per-label notes, confidence-threshold year estimation, and vision-LLM versus OCR-text cost-reduction iterations.
- Reduced manual review load by adding a per-label confidence pipeline that auto-approves high-confidence extractions and routes only uncertain ones to humans, and an OCR cleaning step that normalized raw AWS Textract output for cleaner parsing.
- Closed a class of cross-tenant data-exposure risks by leading a multi-tenancy hardening pass — scoping previously unscoped repository queries by tenant and making tenant scoping mandatory on lookups project-wide.
- Improved throughput on large documents by eliminating recurring N+1 query patterns (bulk existence checks, vectorized NumPy batch matching, batched commits) and fixing crashes in the person-record and embedding steps.
- Migrated event, location, and family-role types from string enums to integer columns with idempotent Alembic migrations and orphaned-value mapping; added composite and fuzzy-name trigram indexes.
- Cut regression risk by introducing the first test coverage for several modules, and shortened review turnaround with Docker deployment and an automated AI PR-review workflow in Bitbucket Pipelines.

Stack: Python | FastAPI | SQLAlchemy 2.0 | Alembic | PostgreSQL | pgvector | sentence-transformers | AWS Textract | OpenAI | Dagster | pytest | Docker

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL

AI: sentence-transformers, Ultralytics YOLOv8, NumPy, OpenAI API, LLM and vision-LLM prompting, OCR (AWS Textract)

Backend & Data: FastAPI, SQLAlchemy 2.0, Alembic, PostgreSQL, pgvector, MySQL, MongoDB, Firebase, Supabase, Prisma ORM, Node.js, Express, Spring Boot, .NET

MLOps & Tools: Docker, Dagster, MLflow, Git, GitHub, Bitbucket Pipelines, pytest, Postman, Jira, Figma, ClickUp

Frontend: React, Next.js, React Native, HTML5, CSS, Tailwind CSS, Chakra UI

PROJECTS

PointEdge — Cross-Platform POS System 06/2024 – 06/2025

- Contributed to a retail POS platform featuring role-based authentication, inventory, customer, and payment management, owning the Shift Management module end to end - data workflows for scheduling, real-time attendance, and reporting.
- Ensured data accuracy across backend integration points and collaborated in an Agile Scrum team with version control. **Stack:** React | Spring Boot | MySQL | GitHub | Figma | ClickUp

Skill Swap — Peer-to-Peer Skill Exchange Platform 2026

- Built a full-stack platform for negotiating one-to-one skill swaps, with the proposal engine as a server-enforced state machine validating skill ownership on offers and counter-offers, credentials auth (NextAuth v5, bcrypt, JWT sessions), and every mutation as a typed Server Action with cache revalidation. **Stack:** Next.js 16 | React 19 | TypeScript | PostgreSQL | Prisma 7 | NextAuth v5 | Tailwind CSS 4

Solo Runner — Self-Training Running Track System 06/2023 – 06/2024

- Built a mobile-responsive web app integrating ESP32-CAM and LDR sensors for automated race capture, with real-time Firebase sync, secure remote access, interactive playback, and performance graphs for athlete training. **Stack:** JavaScript | HTML | CSS | C++ | Firebase

EDUCATION

BSc (Hons) in Information Technology — University of Moratuwa 05/2023 – 2027 (expected)

G.C.E. Advanced Level, Physical Science — Richmond College, Galle | 2A1B 2021

G.C.E. Ordinary Level, English Medium — Richmond College, Galle 2017

CERTIFICATIONS

Claude Certified Architect (CCAF) — Anthropic 2026

ACHIEVEMENTS & PARTICIPATION

Participated in the CodeX HackerRank Competition (CSE) and FIT CodeRush (INTECS), University of Moratuwa, 2025.

EXTRA-CURRICULAR & VOLUNTEERING

- Teaching volunteer with Sansaka Sansadaya and the Rotaract Club, University of Moratuwa.
- Represented Richmond College in cricket and athletics, and served in its Leo Club.

ADDITIONAL

Soft skills: Leadership, Teamwork, Adaptability, Communication, Time management, Problem-solving

Languages: Sinhala (native), English

REFERENCES

Mr. M. D. N. Chandrasiri — Lecturer, Faculty of Information Technology, University of Moratuwa
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